

Near Real-Time Customer Call Sentiment Monitoring Pipeline

1. Data Sources

Two source systems feed the pipeline. Call events arrive as a continuous JSON stream; customer reference data is loaded in batch and kept current via Change Data Capture (CDC).

Source	Type	Description
Call Data	Streaming (JSON)	Generated via Python script; contains nested conversation payloads and rich metadata fields.
Customer Data	Batch + CDC (CSV)	Contains operation field (INSERT / UPDATE / DELETE) and update_ts for CDC simulation.

2. Architecture

The pipeline is built on Databricks Lakeflow Declarative Pipelines following the Medallion Architecture pattern:

Bronze → Raw ingestion with zero business logic, preserving source fidelity.
Silver → Cleansed, deduplicated, and enriched records ready for analytical joins.
Gold → Aggregated KPI datasets optimized for BI dashboards and alerting.

3. Bronze Layer — Raw Ingestion

The Bronze layer captures data exactly as it arrives, adding only operational metadata. No transformations are applied, preserving a complete audit trail.

Table	Type	Details
bronze_calls	Streaming	Ingests JSON via Auto Loader; appends source_file and ingestion_timestamp metadata.
bronze_customers	Streaming	Ingests CSV via read_files with full CDC field support.

4. Silver Layer — Cleansed & Enriched

Call Data Processing

- Convert raw epoch values to a structured event_ts timestamp.
- Apply event-time watermarking to tolerate late-arriving records.
- Deduplicate on composite key (call_id, event_ts) to eliminate reprocessing duplicates.
- Flatten nested JSON arrays and structs into columnar format.

Customer Data Processing

- Filter and quarantine corrupt or malformed records before downstream use.
- Apply CDC operation field to synchronize inserts, updates, and deletes.
- Implement SCD Type 2 via materialized view to preserve the full history of customer attribute changes (e.g., city, subscription tier).

Example — SCD Type 2 history for customer U101:

customer_id	city	start_date	end_date
U101	Delhi	2024-01-01	2025-01-01
U101	Mumbai	2025-01-02	NULL (active)

5. Gold Layer — Business KPIs

The Gold layer exposes five curated KPI datasets consumed directly by dashboards and alerting workflows:

KPI	Description
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Daily Sentiment Distribution	Tracks the ratio of positive to negative call outcomes each day, enabling trend detection.
Daily Call Volume	Aggregates total calls per day to identify traffic peaks and operational patterns.
High-Risk Customers	Flags customers with repeated negative interactions for proactive support escalation.
Region-wise Sentiment	Breaks down sentiment scores by city to reveal geographic service disparities.
Subscription-based Sentiment	Segments sentiment by subscription tier to uncover plan-specific satisfaction gaps.

6. Key Engineering Features

- Streaming ingestion via Databricks Auto Loader with automatic schema detection.
- Schema enforcement and seamless evolution handling to accommodate source changes without pipeline restarts.
- Watermark-based late-data handling anchored to event time, not ingestion time.
- CDC simulation using the operation column for lightweight change tracking without a dedicated CDC tool.
- SCD Type 2 materialized view for full historical auditability of customer attributes.
- Operational metadata columns (source_file, ingestion_timestamp) for lineage tracking and debugging.

7. DAG



8 Expected Outcome

The pipeline produces **clean, reliable, and analysis-ready data** by ingesting streaming call data and batch customer data. It ensures **data quality, deduplication, and handling of late-arriving data** using event-time processing.

Customer data is managed using **SCD Type 2**, enabling **historical tracking of changes** such as city and subscription type.

The final output includes **business KPIs** such as daily call volume, sentiment distribution, high-risk customers, and customer segmentation by region and subscription type. The pipeline supports **incremental processing**, ensuring that insights are continuously updated as new data arrives.